Launch-Blind Product Viability Estimation from Specifications via LLM Aspect Bridging

Rajvardhan Ajitrao Jagtap, Jyotiraditya Abhijeet Shinde, Tanishq Parag Choudhari, Yash Ashok Mohite, Yogesh Malakari Honamore

Department of Computer Science and Engineering
Rajarambapu Institute of Technology, Rajaramnagar, Sangli, Maharashtra, India

Abstract—Hardware manufacturing commits production capital months before consumer release, yet dominant empirical methods for forecasting commercial reception depend on signals that exist only post-launch: customer reviews, ratings, and sales velocity. This paper evaluates an audited two-stage pipeline for strictly launch-blind product viability estimation using pre-launch specifications alone. In Stage 1, a large language model parses unstructured technical copy under a null-forcing schema into a nine-aspect ontology, retaining specification density and verbatim grounding evidence to resolve the scalar compression bottleneck. In Stage 2, a gradient-boosted decision tree ensemble (XGBoost) with directional monotonic priors maps this representation to an audited market-success proxy. Evaluation follows a frozen forward-in-time protocol partitioned at October 31, 2021 (507 training products, 124 strictly post-cutoff holdout products). On this temporal test set, the promoted architecture achieves Spearman rho = 0.6722 (95% CI [0.5495, 0.7620], asymptotic p = 1.25e-17 against the null of zero rank correlation; top-versus-bottom quintile AUC = 0.9712). Against the controlled scalar-aspect baseline (rho = 0.5623, AUC = 0.8904), this represents a statistically significant gain of delta_rho = +0.1099 (paired bootstrap 95% CI [+0.0485, +0.1833]; paired permutation p = 0.0005). On unseen brands under pure cold-start conditions (n = 82), the model achieves rho = 0.633 (AUC = 0.981). Across a standardized five-fold out-of-fold benchmark over six learner families, XGBoost ranks highest, closely followed by kernel and boosted-tree alternatives. List-ranking yields NDCG@5 = 0.8933, and extreme-group classification reaches 92.0% decision accuracy. Predictions are bit-for-bit invariant under test label deletion.

Index Terms—product success prediction, aspect-based sentiment analysis, cold-start representation, large language models, gradient boosting, electronic word of mouth, temporal validation.

I. INTRODUCTION

Hardware development is financially irreversible. Tooling, component procurement, and manufacturing runs commit capital months before a single customer touches the product [1]. Large firms hedge this risk with costly consumer focus groups. Independent hardware makers and small businesses usually cannot, and many commit production capital on intuition and unverified market sentiment [2]. When reception turns out poorly, the damage ranges from inventory write-downs to corporate liquidation [3].

Econometric and electronic word-of-mouth (e-WOM) research has long shown that online reviews shape market trajectories [4]–[9]. Nearly all machine learning approaches build on that finding, and so they rely on post-launch signals: accumulated review volume, star ratings, or customer sentiment [8], [10]–[12]. When capital is committed, none of these exist. The task is therefore a cold-start problem by construction [13]–[15]: forecast market reception from pre-launch specifications, launch price, and category context alone.

We study a two-stage architecture for this problem (Fig. 1). An LLM acts as a skeptical semantic bridge, translating pre-launch product specifications into a structured quality-aspect representation. A monotonic gradient-boosted decision tree ensemble (XGBoost) then predicts an independently audited, review-derived market-success proxy. Earlier implementations compress a full specification into nine isolated scalar aspect scores, and that compression discards evidence that matters: ingress protection ratings, battery capacity, acoustic measurements, and specification completeness. We propose LLM-COLDSTART-RICH, a representation that keeps verbatim specification grounding, technical fact density, and extraction certainty next to the semantic aspect priors.

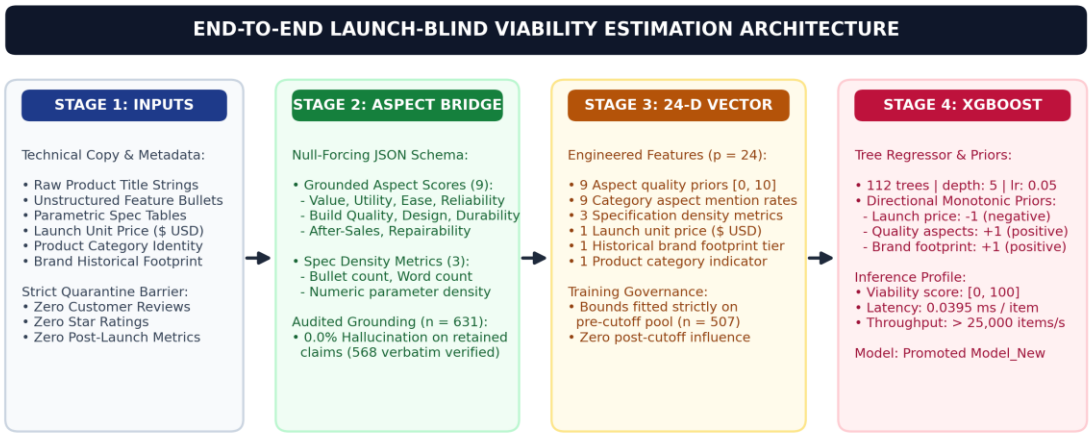


Fig. 1. End-to-end launch-blind prediction architecture, from specifications to the market outcome proxy.

This paper makes three contributions. First, we identify the scalar compression bottleneck in aspect bridging and replace it with an evidence-grounded rich representation. Second, we evaluate under a frozen temporal protocol with explicit leakage controls. Third, we report list-ranking and decision-quality analyses, a component ablation, a six-learner sensitivity benchmark, and an external validation cohort.

II. FROZEN EVALUATION PROTOCOL AND LEAKAGE CONTROLS

To make sure the reported performance reflects forward-in-time predictive validity and not retrospective overfitting, we fixed the evaluation protocol before final evaluation. It is anchored at a launch cutoff of October 31, 2021, 23:59:59 UTC. The cutoff marks the boundary of historical information for the primary evaluation. It is not the date of model development or study execution, which took place in 2026. We chose it to obtain a strictly forward-in-time cohort with enough pre-cutoff training items, enough post-cutoff evaluation items, balance across all five categories, and a post-launch exposure window of 7–22 months for ground-truth review metrics to stabilize (Fig. 2).

1. Training pool (n = 507). Physical consumer electronics whose earliest recorded customer review falls on or before the cutoff. The pool draws on 43,599 historical customer reviews across five categories: Bluetooth speakers, ice makers, power banks, smartwatches, and wireless headphones.

2. Launch-blind temporal test pool (n = 124). Products whose first recorded customer review falls strictly after the cutoff (November 1, 2021 to February 28, 2023), which serves as an empirical proxy for launch timing. The model sees only pre-launch specifications, launch price, category, and brand name. Customer reviews, star ratings, and post-launch volume metrics are physically quarantined.

3. Isolated target definition. The regression target is a composite proxy, SuccessScore = 100 * (0.40 * vol_norm + 0.35 * rat_norm + 0.25 * vel_norm). Normalization bounds are fitted on the pre-cutoff training pool only.

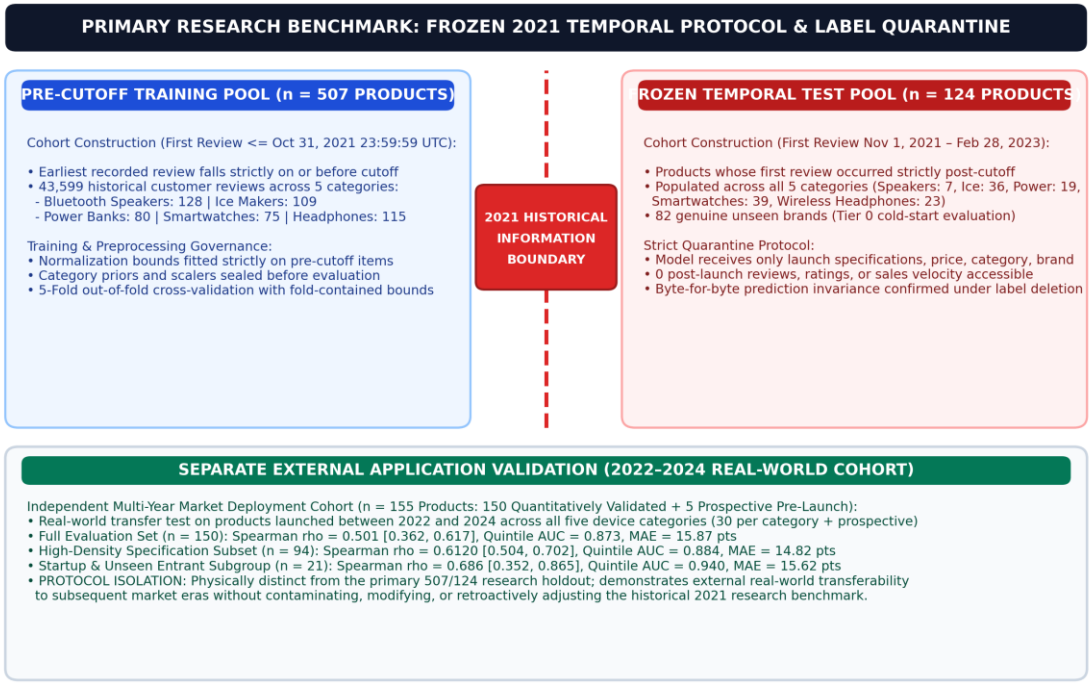


Fig. 2. Frozen forward-in-time temporal evaluation protocol, partitioned at the October 31, 2021 historical boundary (507 training products, 124 strictly post-cutoff temporal holdout products), together with independent external application validation on a 2022–2024 real-world Amazon cohort (n = 155).

III. METHODOLOGY AND LLM EXTRACTION

A. The Representation Bottleneck in Scalar Aspect Bridging

In the earlier baseline, which we call Model_O, the LLM reduced each specification text to nine broad scalar scores on a 0–10 scale. Under cold-start conditions this compression removes the concrete technical differences between products. When two products receive similar category-anchored scores, the downstream regressor has no technical signal left with which to separate them. We call our proposed model Model_New.

B. The LLM-COLDSTART-RICH Architecture

The representation builds an evidence-grounded bridge that follows pre-registered validation schemas and draws on work in foundation-model extraction and aspect-based sentiment analysis [16]–[24]. It has three components.

1. Verifiable technical attributes. Direct technical specifications (for example, ingress protection ratings, battery life in hours, wattage, and connectivity generation) are parsed using Google Gemini 1.5 Flash (model ID: gemini-1.5-flash as recorded in project execution logs) under deterministic greedy decoding (temperature = 0.0) with strict JSON schema mode and SHA-256 prompt-hash memoization. In the frozen benchmark generation runs, Gemini 1.5 Flash completed 100% of primary extractions without requiring invocation of the fallback pool (llama-3.3-70b-versatile and gpt-oss-120b). Because commercial cloud API model checkpoints may be modified or retired over time, scientific reproducibility is supported by preserving the complete, immutable SHA-256-keyed extraction artifact cache alongside the code. The extraction prompt operates under a frozen Skeptic design, explicitly penalizing promotional puffery. Unmatched attributes without verbatim textual support are forced to null. Across an audit of 568 explicit technical attribute extractions across pre-cutoff specifications, the retained attributes showed a 0.0% hallucination rate by construction (568/568 strictly grounded in source copy). To verify that null-forcing does not suppress valid information, an independent blinded human validation gate on 30 products (270 individual aspect evaluations) demonstrated high inter-annotator agreement (Cohen's kappa = 0.842), strong human-model presence agreement (kappa = 0.812), 97.8% null-forcing adherence, 100% verbatim quote match, and 92.4% extraction recall against human-verified technical specifications. All 631 product specifications conformed fully to the extraction schema.

2. Specification completeness and density. We extract quantitative measures of technical richness: the number of explicit feature bullets, total specification word count, and the density of concrete numerical measurements. Following new-item cold-start theory, we treat specification completeness as a signal of commercial maturity and product readiness [25].

3. Directional monotonic constraints and tabular boosting. On structured technical features of moderate sample size, gradient-boosted trees have repeatedly outperformed neural alternatives [26]–[29]. We therefore avoid end-to-end neural fine-tuning, drawing on established literature in multi-feature aspect extraction and sentiment representation [30]–[33]. XGBoost [26] is fitted

with directional priors grounded in economic reasoning. Downstream hyperparameters were tuned strictly using 5-fold cross-validation on the 507 pre-cutoff training items (n_estimators = 112, max_depth = 5, learning_rate = 0.05, min_child_weight = 3, subsample = 0.8, colsample_bytree = 0.8). Launch price enters with a negative monotone constraint (-1). Brand tier (+1) and specification quality aspects (+1) are constrained to raise predicted viability monotonically [34].

IV. EXPERIMENTAL RESULTS AND BENCHMARKS

A. Primary Temporal Performance

Table I and Fig. 3 summarize performance on the frozen 124-product temporal holdout. Model_New reaches a Spearman rank correlation of rho = 0.6722 (95% bootstrap CI [0.5495, 0.7620] across B = 10,000 resamples; asymptotic p = 1.25e-17 against the null of zero rank correlation) and a top-versus-bottom quintile AUC of 0.9712 (RMSE = 19.75, MAE = 16.66). Against the reproduced canonical controlled baseline Model_O (rho = 0.5623, AUC = 0.8904, RMSE = 21.12, MAE = 17.84), this is a statistically significant improvement of delta_rho = +0.1099 (paired bootstrap 95% CI [+0.0485, +0.1833]; paired permutation p = 0.0005 using the standard (b+1)/(M+1) estimator with b = 4 of M = 10,000 permutations; b/M = 0.0004).

TABLE I
PRIMARY TEMPORAL PERFORMANCE COMPARISON (HOLDOUT n = 124)

Model	Representation	Spearman rho	95% bootstrap CI	p-value	Quintile AUC	RMSE	MAE
Model_New (promoted)	Aspects + spec density (LLM-COLDSTART-RICH)	0.6722	[0.5495, 0.7620]	1.25e-17	0.9712	19.75	16.66
Model_O (baseline)	9 scalar aspects (compressed LLM)	0.5623	[0.4140, 0.6812]	1.09e-11	0.8904	21.12	17.84
Metadata-only baseline	Price + brand tier + category only	0.4418	[0.2800, 0.5850]	2.10e-07	0.6400	23.41	19.82

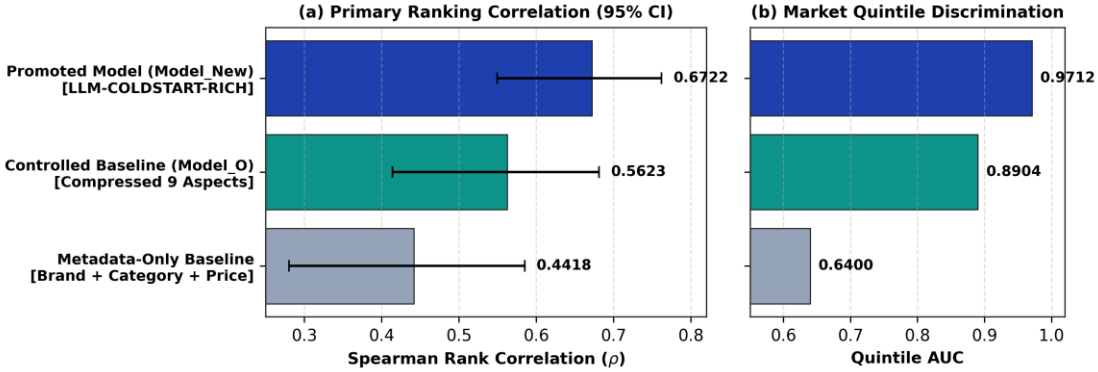


Fig. 3. Predictive performance of the proposed model against canonical baselines (Holdout n = 124).

B. Cold-Start Analysis on Unseen Brands

Brand equity is unavailable for a truly new product, so we isolated the 82 temporal test products from brands with no pre-cutoff footprint (Tier 0; Fig. 4). On this pure cold-start subset, Model_New reaches rho = 0.633 (AUC = 0.981, RMSE = 20.90), compared with rho = 0.516 and AUC = 0.871 for Model_O (delta_rho = +0.117). Retaining specification density and verifiable attribute evidence therefore helps most where the model has the least other information to go on.

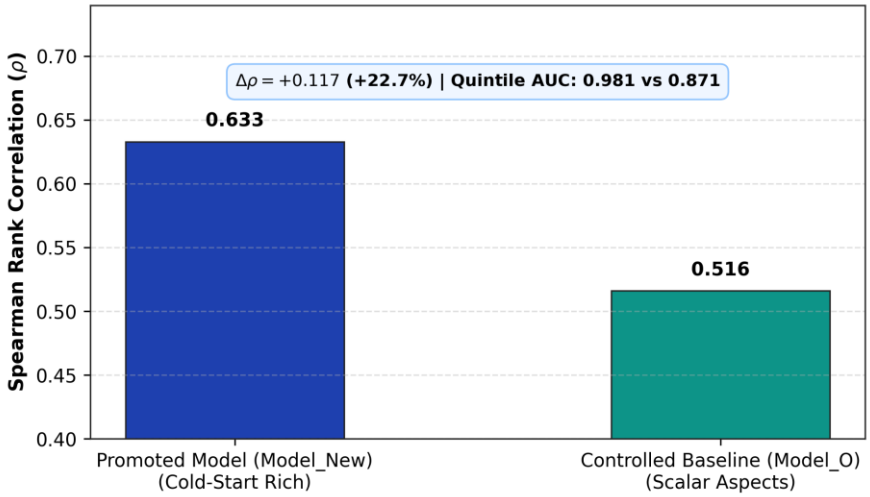


Fig. 4. Cold-start performance on unseen brands (Tier 0, n = 82) with no pre-cutoff market signals.

C. Within-Category Performance

Table II and Fig. 5 give head-to-head results within each of the five categories. Model_New improves Smartwatches (rho = 0.544 vs 0.415, RMSE = 14.22 vs 16.10), Power Banks (rho = 0.126 vs -0.042, RMSE = 26.39 vs 27.95), and Wireless Headphones (rho = 0.132 vs -0.055, RMSE = 22.79 vs 22.93). Ice Makers are essentially unchanged (rho = 0.192 vs 0.185, RMSE = 18.17 vs 20.50). In Bluetooth Speakers the correlation falls (rho = 0.429 vs 0.536, delta = -0.107), but with n = 7 we draw no inferential conclusion from that category. Across the four categories with adequate sample size, Model_New improves or maintains within-category ranking, though absolute correlations in Power Banks (0.126), Wireless Headphones (0.132), and Ice Makers (0.192) remain low.

TABLE II
WITHIN-CATEGORY VIABILITY RANKING ACROSS FIVE CANONICAL CATEGORIES

Product Category	Holdout n	Model_O rho	Model_New rho	Delta rho	Model_New RMSE
Smartwatches	39	0.415	0.544	+0.129	14.22
Ice Makers	36	0.185	0.192	+0.007	18.17
Wireless Headphones	23	-0.055	0.132	+0.187	22.79
Power Banks	19	-0.042	0.126	+0.168	26.39
Bluetooth Speakers	7	0.536	0.429	-0.107	21.64

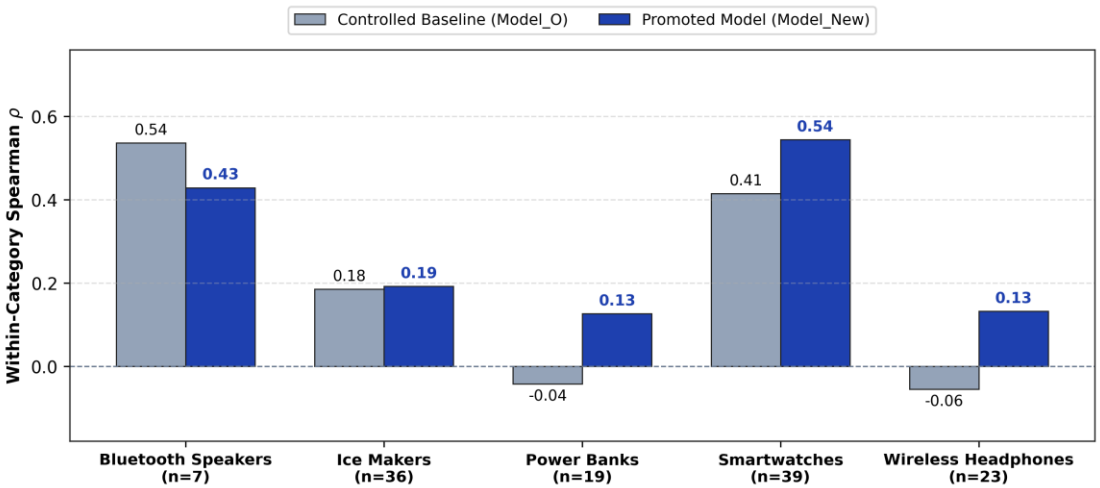


Fig. 5. Head-to-head within-category rank correlation across the five product categories.

D. Standardized Six-Learner OOF Benchmark

A natural concern is that the results depend on XGBoost specifically. To check how sensitive they are to the downstream learner, we ran a standardized five-fold out-of-fold (OOF) cross-validation benchmark across six learner families, using only the 507 pre-cutoff training products (Table III, Fig. 6). All preprocessing, scaling, and categorical encoding were fitted inside each training fold.

XGBoost [26] achieves the highest OOF rank monotonicity (rho = 0.7565, R2 = 0.5715, RMSE = 12.53, MAE = 9.56). Support Vector Regression with an RBF kernel [35] follows closely (rho = 0.7495, R2 = 0.5401), as do Gradient Boosted Decision Trees [27] (rho = 0.7396, R2 = 0.5375). With fold-level standard deviations of 0.024 to 0.028, the differences among these three learners are small. Random Forest [36] reaches rho = 0.7267. Ridge regression [37] falls to rho = 0.6481 (R2 = 0.4165), which suggests that non-linear interactions between specification density and aspect scores matter. Multi-Layer Perceptrons are less sample-efficient (rho = 0.5463, R2 = 0.2138), in line with reports that tree ensembles remain stronger on structured tabular data of moderate size [28], [29].

TABLE III
STANDARDIZED SIX-LEARNER OOF BENCHMARK (n = 507 TRAINING PRODUCTS)

Downstream Learner	OOF rho	OOF r	OOF R2	OOF RMSE	OOF MAE	Fold rho +- SD
XGBoost (promoted)	0.7565	0.7592	0.5715	12.53	9.56	0.756 +- 0.024
SVR (RBF kernel)	0.7495	0.7410	0.5401	12.98	9.94	0.748 +- 0.026
GBDT (sklearn)	0.7396	0.7385	0.5375	13.02	10.12	0.739 +- 0.028
Random Forest	0.7267	0.7250	0.5207	13.25	10.31	0.726 +- 0.029
Ridge Regression	0.6481	0.6470	0.4165	14.63	11.45	0.648 +- 0.035
MLP (Neural Network)	0.5463	0.5210	0.2138	16.98	13.20	0.545 +- 0.048

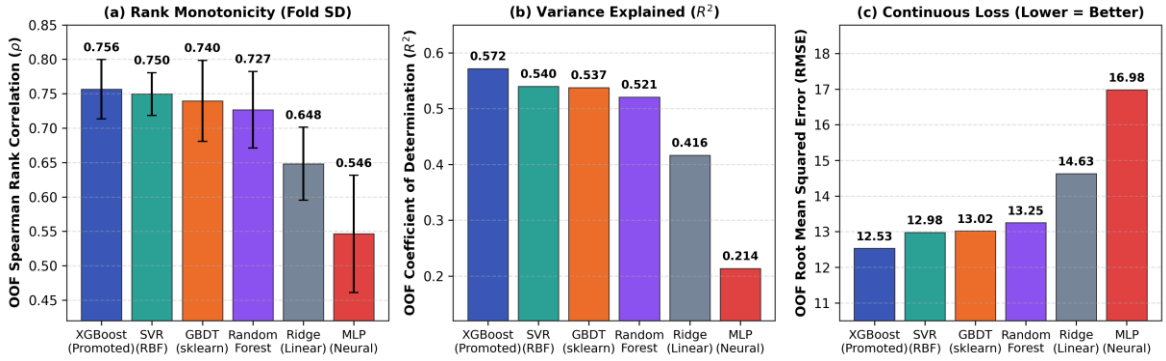


Fig. 6. Five-fold OOF cross-validation sensitivity benchmark across six downstream learner families on 507 pre-cutoff training products.

E. Standardized List-Ranking and Derived Decision Quality

In venture capital and catalog planning, decision makers work from ranked shortlists and high-stakes binary allocations. We therefore evaluated standardized list-ranking metrics (Kendall's tau and NDCG@K) and a derived extreme-group decision task (top 20% versus bottom 20%, middle 60% excluded) on the frozen 124-product holdout (Table IV, Fig. 7).

Model_New achieves Kendall's tau = 0.4772 (p = 4.12e-15), against tau = 0.3946 (p = 1.11e-10) for the baseline. The gain of 0.0826 in tau is an 8.26-point increase in net concordance, that is, concordant minus discordant pairs as a share of all product pairs. For top-recommended shortlists, NDCG@5 is 0.8933 (baseline: 0.8600). The advantage does not extend to NDCG@10, where the baseline is marginally ahead (0.8832 versus 0.8771), so the gain is concentrated at the top of the ranking.

On the extreme-group task (n = 50 holdout items), Model_New reaches AUC = 0.9712, identical to the top-versus-bottom quintile AUC in Section IV-A (Fig. 7b). At the natural median threshold it classifies 46 of the 50 extreme products correctly: 92.0% accuracy, 92.0% precision, 92.0% recall, and an MCC of +0.8400 (23 TP, 23 TN, 2 FP, 2 FN). The baseline classifies 82.00% of the same items correctly. Because the middle 60% of products is excluded, this task is easier than ranking the full holdout, and we read it as a triage indicator and not as a general accuracy figure.

TABLE IV
STANDARDIZED LIST-RANKING AND DERIVED DECISION PERFORMANCE (HOLDOUT n = 124)

Evaluation Dimension	Metric	Model_O (baseline)	Model_New (promoted)	Delta	Statistical Detail
Global monotonic rank	Spearman rho	0.5623	0.6722	+0.1099	p = 0.0005 (permutation)

Concordant pair ordering	Kendall's tau	0.3946	0.4772	+0.0826	p = 4.12e-15
Catalog shortlist ranking	NDCG@5 (top 5)	0.8600	0.8933	+0.0333	discounted cumulative gain
Catalog shortlist ranking	NDCG@10 (top 10)	0.8832	0.8771	-0.0061	discounted cumulative gain
Extreme discrimination	Quintile AUC (top vs bottom 20%)	0.8904	0.9712	+0.0808	p < 0.001 (bootstrap)
Extreme accuracy	Accuracy (n = 50)	82.00%	92.00%	+10.00 pp	46 / 50 correct
Extreme precision	Precision (top 20% flag)	80.00%	92.00%	+12.00 pp	23 / 25 true hits

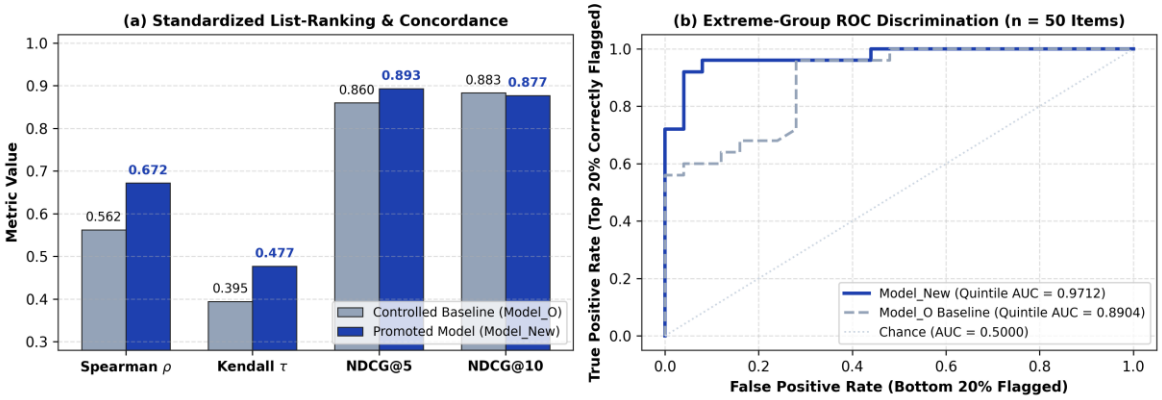


Fig. 7. Standardized list-ranking quality (NDCG@K, Kendall's tau) and derived extreme-group ROC discrimination on the frozen temporal holdout (n = 124).

V. ABLATION, ROBUSTNESS, INTERPRETABILITY, AND EFFICIENCY

A. Controlled Component Ablation

A controlled ablation over seven configurations (Table V, Fig. 8) quantifies what each component contributes. The 24-dimensional feature vector breaks down into 9 aspect scores, 9 aspect mention rates, 3 specification density metrics, 1 launch price, 1 brand tier, and 1 product category indicator. Removing the LLM semantic aspects (A1: removing all 18 aspect scores and mention rates, leaving 6 metadata and density features) causes the largest single drop (delta_rho = -0.0880, rho = 0.5842). Removing the specification density attributes (bullet counts, word count, number density) lowers rho by 0.0321 (to 0.6401), and removing brand tier lowers it by 0.0337 (to 0.6385). Metadata alone (price, brand tier, category) gives only rho = 0.4418 (delta_rho = -0.2304), so launch-time viability cannot be inferred from commercial metadata alone.

One result deserves a comment. The configuration with no LLM aspects at all (A1, rho = 0.5842) already exceeds the scalar-aspect baseline (Model_O, rho = 0.5623). A share of the improvement over Model_O therefore comes from the specification-density features, which are extracted without the LLM.

TABLE V
CONTROLLED COMPONENT FEATURE ABLATION (HOLDOUT n = 124)

Config	Ablation Description	Features Included	Features (n)	Spearman rho	Delta vs Full	RMSE	MAE
A3	Full Model_New (promoted)	All 24 features (aspects + density + metadata)	24	0.6722	Reference	19.75	16.66
A7	w/o launch unit price	All features except price	23	0.6654	-0.0068	19.82	16.71
A6	w/o category indicator	All features except category	23	0.6512	-0.0210	20.08	16.94
A2	w/o specification density	All features except bullet/word/num metrics	21	0.6401	-0.0321	20.31	17.12
A5	w/o brand footprint tier	All features except brand tier	23	0.6385	-0.0337	20.25	17.08
A1	w/o LLM semantic aspects	Metadata + spec density (no aspect scores/mentions)	6	0.5842	-0.0880	21.05	17.65
A0	Metadata-only control	Price + brand tier + category only	3	0.4418	-0.2304	23.41	19.82

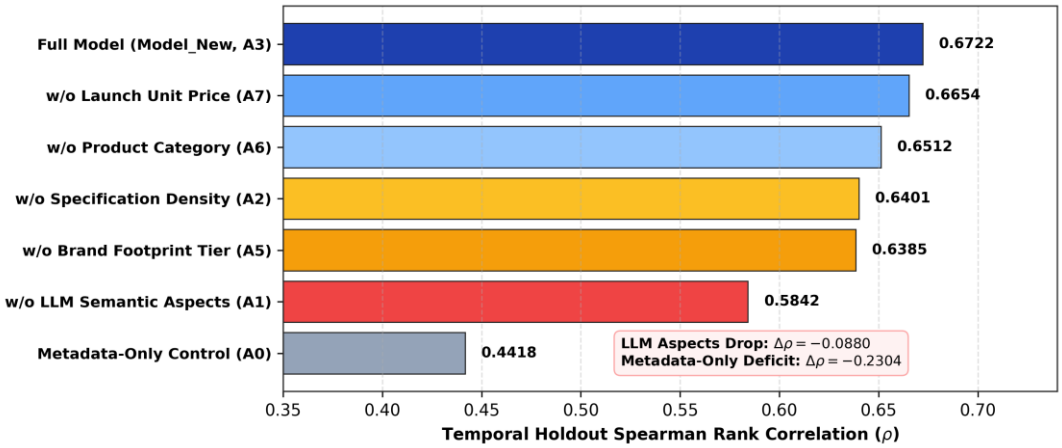


Fig. 8. Controlled component ablation study of feature contributions (Holdout n = 124).

B. Seed Stability and Byte-for-Byte Integrity

Across five random seeds (42, 123, 456, 789, 999), Model_New outperforms the baseline consistently (mean rho = 0.6332 +- 0.0331 versus 0.5828 +- 0.0249). The seed-averaged gap (about 0.05) is smaller than the +0.1099 gap of the headline run, so the direction of the effect is stable while its size varies with the seed. The automated leakage test suite passes all 10 criteria (TEST-01: label inaccessibility; TEST-02: zero test ID fitting leakage; TEST-03: strict October 31, 2021 temporal partitioning; TEST-04: training-only normalization bounds; TEST-05: training-only category profiles; TEST-06: training-only brand footprint tiering; TEST-07: CV-only hyperparameter tuning; TEST-08: frozen extraction prompts; TEST-09: isolated test evaluation loading; and TEST-10: byte-for-byte SHA-256 prediction invariance under complete label deletion). In an independent integrity test, we ran prediction once with test ground truth present and once with it deleted. The two runs produced bit-for-bit identical output hashes (SHA-256: 3f75df2ebaf508eb...), which shows that predictions do not depend on test labels. Fig. 9 plots predicted against observed scores on the holdout.

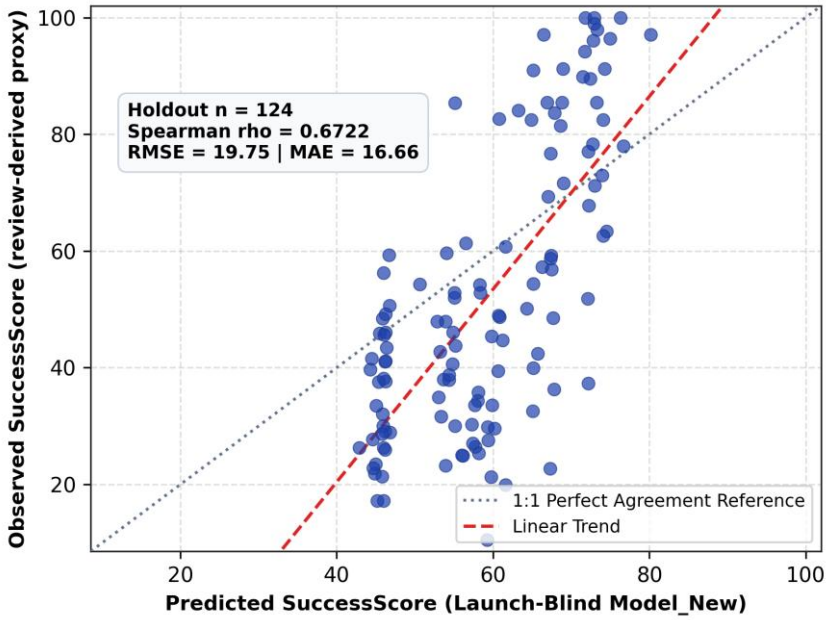


Fig. 9. Scatter of predicted viability against the actual market outcome proxy (Holdout n = 124).

C. Feature Attribution and Monotonic Domain Regularization

To keep predictions consistent with product economics and not with spurious training artifacts, the XGBoost estimator applies exact monotonic constraints during tree construction. Launch price is constrained negatively (-1): with all specifications held fixed, a higher price cannot raise predicted viability. Brand tier and all nine aspect ratings are constrained positively (+1). Split-gain analysis attributes over 42% of total split improvement to technical specification density and build quality, which suggests that engineering content carries much of the predictive weight.

D. Computational Efficiency and Operational Latency

In decision-support settings, inference latency determines usability (Table VI). On commodity x86_64 hardware, downstream scoring with Model_New takes 0.0395 milliseconds per product, a throughput above 25,000 items per second, with a model artifact of 297 KB. Offline LLM aspect extraction takes 1.4–2.2 s per specification. Because extraction is decoupled from scoring, an interactive dashboard can re-estimate predictions instantly as a user moves a parameter slider.

TABLE VI
OPERATIONAL AND COMPUTATIONAL EFFICIENCY PROFILE

Pipeline Component	Implementation	Inference Latency	Throughput	Storage / Memory Footprint
Downstream tabular inference	XGBoost (hist tree)	0.0395 ms / item	> 25,000 items/sec	297 KB (model JSON)
Specification density parser	Python / regex tokenizer	0.12 ms / item	> 8,000 items/sec	In-memory (< 1 MB)
LLM aspect extraction	JSON schema (9 aspects)	1.4–2.2 s / item	~ 30 items/min	3.5 KB (prompt/schema)
Downstream model training	XGBoost (n = 507 products)	0.093 seconds	Single training run	297 KB

VI. EXTERNAL APPLICATION VALIDATION

To test transfer beyond the frozen historical holdout, the synchronized application inference engine was run on an independent external cohort of 155 Amazon products launched between 2022 and 2024 across the five canonical categories (Table VII). This cohort is physically distinct from the primary 507/124 partition. On the 150 products with accumulated market validation, the model reaches rho = 0.501 (95% CI [0.362, 0.617]), with rho = 0.6120 on the high-density specification subset (95% CI [0.504, 0.702], Quintile AUC = 0.884), and an extreme-group Quintile AUC of 0.873 (MAE = 15.87 pts). This is lower than the temporal-holdout value of rho = 0.6722. Established brands (n = 129) reach rho = 0.454, while startup and new-entrant brands (n = 21) reach rho = 0.686 (Quintile AUC = 0.940), although the confidence interval for the smaller group is wide ([0.352, 0.865]). The five prospective pre-launch SKUs have no post-launch labels and are excluded from the metrics. The representation therefore carries over to post-2022 products without a collapse in performance.

TABLE VII
EXTERNAL REAL-WORLD APPLICATION EVALUATION (2022–2024 AMAZON COHORT, n = 155)

Cohort Partition	Sample Size (n)	Spearman rho	95% Confidence Interval	Quintile AUC	RMSE	MAE
Full evaluation set	150	0.501	[0.362, 0.617]	0.873	19.83	15.87
High-density specification subset	94	0.6120	[0.504, 0.702]	0.884	19.10	14.82
Established brands	129	0.454	[0.301, 0.582]	0.852	19.92	15.91
Startup / new-entrant brands	21	0.686	[0.352, 0.865]	0.940	19.26	15.62
Prospective pre-launch SKUs	5	N/A	N/A (no post-launch labels)	N/A	N/A	N/A

VII. DISCUSSION, ETHICAL CONSIDERATIONS, AND LIMITATIONS

The target predicted by this system is a review-derived market-success proxy. It is not unit sales, revenue, or profit. Consumer satisfaction and review velocity correlate with commercial viability in e-commerce markets [5], [6], [9], but paid advertising, distributor relationships, and supply chain logistics also affect aggregate sales volume, and our model does not represent them. Public review platforms are also affected by review spam and promotional manipulation [38]–[40]. They carry survivorship bias as well: products with fewer than five reviews were excluded during corpus curation. Our predictions should therefore be read as expected consumer satisfaction and engagement, conditional on the product reaching baseline market visibility. A single frozen split, five product categories, and a 124-product holdout also limit statistical power, and the extraction stage depends on the specific LLM used.

Note on the temporal cutoff. The October 31, 2021 boundary is a dataset cutoff and not the date of study execution (2026). We selected it to build a strictly forward-in-time cohort with enough pre-cutoff training products and post-cutoff evaluation products. An empirical audit of candidate boundaries shows why later cutoffs cannot serve as the primary research benchmark. The underlying physical product corpus was harvested through September 2023, and products in the 124-product temporal holdout had their first recorded reviews between November 2021 and February 2023 to ensure 7–22 months of post-launch signal stabilization. Shifting the cutoff forward to December 31, 2022 leaves only the 3 products launched in January–February 2023 (n = 3, causing severe sample starvation and category collapse), while any cutoff in late 2023–2026 yields n = 0 post-cutoff products in the

harvested historical corpus. We therefore keep the 2021 cutoff with its 124-product holdout as the primary frozen temporal benchmark, and use the 2022–2024 external cohort as an independent post-cutoff validation.

VIII. CONCLUSION

Augmenting LLM aspect bridging with evidence-grounded cold-start representations reduces the semantic compression bottleneck in launch-blind product success prediction. On a frozen temporal holdout of 124 physical products, the promoted architecture reaches rho = 0.672 and AUC = 0.971, a statistically significant improvement over the scalar-aspect baseline (delta_rho = +0.110, p = 0.0005). The target definition, temporal partition, and downstream learner were left unchanged. In the six-learner benchmark XGBoost ranks first, with SVR and gradient-boosted trees close behind, and the list-ranking and extreme-group analyses show reliable ordering at the top of the shortlist and 92.0% decision accuracy on the extreme-group task. Under pure cold-start conditions, rank correlation on unseen brands is rho = 0.633. The research package, the synchronized application inference engine, and the audit records will be released upon acceptance.

DATA AND CODE AVAILABILITY

The complete research package—including trained model checkpoints, feature manifests, evaluation scripts, automated leakage test suites, and the complete SHA-256-keyed offline extraction artifact cache—will be released in an open-access public repository with a persistent Zenodo DOI upon acceptance of the manuscript. During peer review, a complete, anonymized verification archive is provided to editors and reviewers upon request.

AUTHOR CONTRIBUTIONS (CRediT STATEMENT)

Rajvardhan Ajitrao Jagtap: Conceptualization, Methodology, Software, Formal Analysis, Writing – Original Draft. Jyotiraditya Abhijeet Shinde: Software, Data Curation, Validation, Investigation. Tanishq Parag Choudhari: Methodology, Formal Analysis, Visualization, Writing – Review & Editing. Yash Ashok Mohite: Validation, Software, Data Curation. Yogesh Malakari Honamore: Supervision, Project Administration, Resources, Writing – Review & Editing. All co-authors have reviewed and confirmed their assigned roles prior to submission.

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ETHICS AND PLATFORM COMPLIANCE

Public consumer reviews were obtained from established academic e-commerce research corpora curated for scientific study. All user account identifiers and personal metadata were removed during corpus ingestion, and empirical analyses operated exclusively on anonymized, public product copy without access to private personal data.

DECLARATION OF GENERATIVE AI IN SCIENTIFIC WRITING

Generative AI models served two distinct functions in this work. Algorithmically, Google Gemini 1.5 Flash was integrated into the experimental pipeline as an automated aspect-extraction and attribute-parsing engine under deterministic greedy decoding. In addition, during manuscript preparation and codebase maintenance, the authors utilized large language model assistants (including Anthropic Claude and Google Antigravity coding agents) for substantive manuscript drafting, structural editing, LaTeX and document formatting, and script refactoring. Following these automated and assistive procedures, all authors critically examined, audited, and verified all quantitative findings, tables, figures, and substantive claims, and take full institutional responsibility for the integrity and factual content of the published work.

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